

ENGR 102 Summer Bridge

Day 14 Instructor Key

List Indexing, Mutation, and Final-State Summaries

20 points • approximately 20-25 minutes

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Boolean-operator convention. Python operators appear as [and](#), [or](#), and [not](#). Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Trace mutation and index-dependent updates.

4 points

```
values = [4, -2, 7, -1]
changed = 0

for index in range(len(values)):
    if values[index] < 0:
        values[index] = 0
        changed = changed + 1
    values[index] = values[index] + index

print(values, changed)
```

Give the list and changed state after every index, then write exact output.

Answer and explanation

Index states are 0 -> [4, -2, 7, -1], 0; 1 -> [4, 1, 7, -1], 1; 2 -> [4, 1, 9, -1], 1; 3 -> [4, 1, 9, 3], 2. Exact output: [4, 1, 9, 3] 2.

Question 2. Repair a loop that changes only its temporary value.

4 points

```
values = [-2, 4, -1]
total = 0

for value in values:
    if value < 0:
        value = 0
    total = total + value

print(values, total)
```

Give actual output, explain why the list does not change, and rewrite the loop header/body to mutate positions.

Answer and explanation

The local name value becomes 0 for negative passes, so total is 4, but no list position is assigned. Actual output: [-2, 4, -1] 4. Repair by looping over `range(len(values))` and assigning `values[index] = 0` when negative.

Question 3. Program construction**12 points**

Write one complete Python program that satisfies every requirement.

- Read threshold as an integer and use readings = [12, -3, 7, 20, 7].
- Loop by index and replace every negative value with 0.
- Using final values, compute total and the count at or above threshold.
- Find the index of the first largest final value without max or index.
- Print Final, Total, At or above, and Largest index on separate labeled lines.

Answer and explanation

```
threshold = int(input())
readings = [12, -3, 7, 20, 7]
total = 0
at_or_above = 0
largest = readings[0]
largest_index = 0

for index in range(len(readings)):
    if readings[index] < 0:
        readings[index] = 0
    value = readings[index]
    total = total + value
    if value >= threshold:
        at_or_above = at_or_above + 1
    if value > largest:
        largest = value
        largest_index = index

print(f"Final: {readings}")
print(f"Total: {total}")
print(f"At or above: {at_or_above}")
print(f"Largest index: {largest_index}")
```

Mutation occurs before all summaries, so total, threshold count, and largest state describe the final list. Strict > preserves the first largest index if a tie occurs.

With threshold 10, the final list is [12, 0, 7, 20, 7], total is 46, at_or_above is 2, and largest_index is 3.

Scoring guidance

2 input/initialization, 2 indexed mutation, 3 final total/count, 2 first-largest tracking, 1 exact output, 2 tests/explanation.